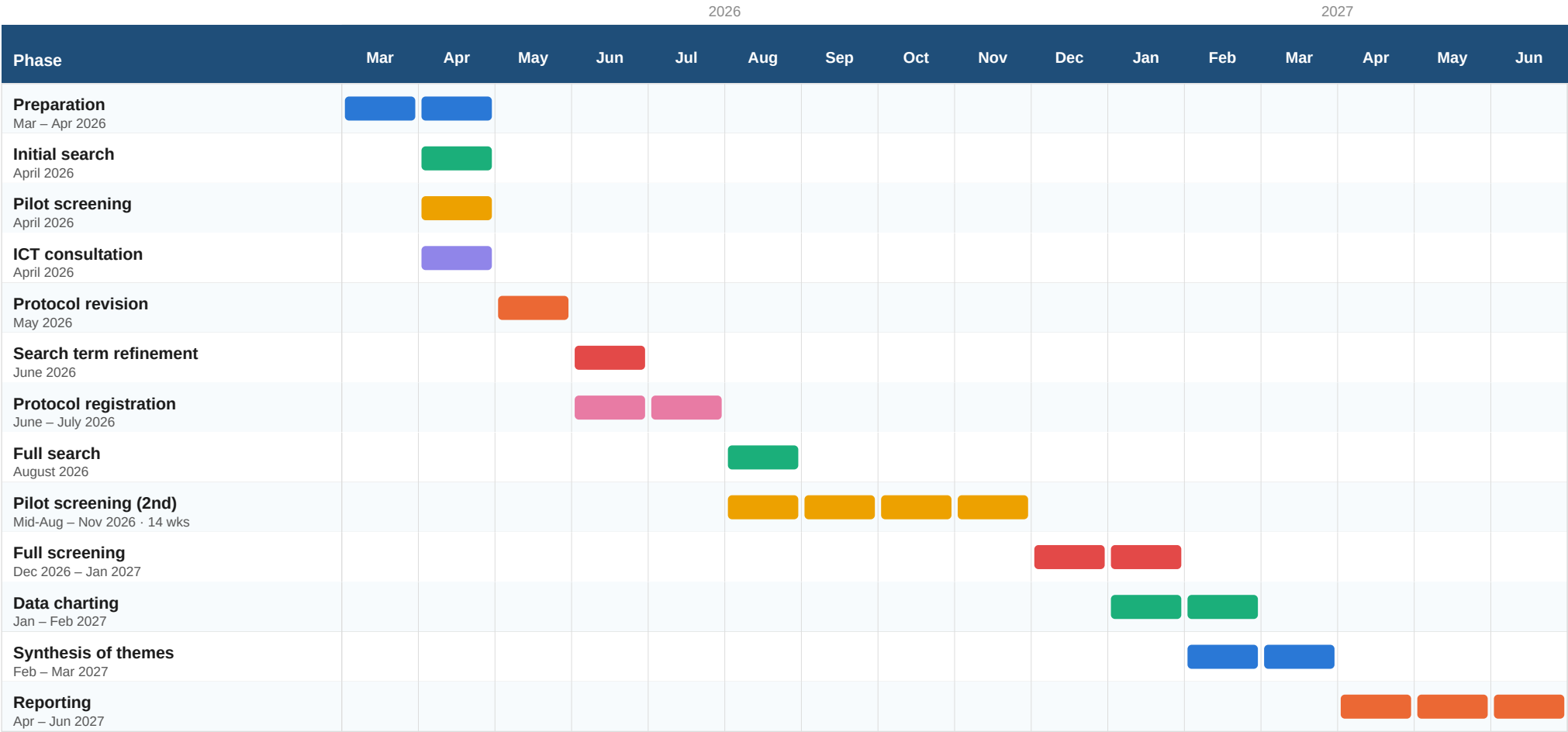


SCOPING REVIEW — PROJECT GANTT CHART | March 2026 – June 2027

- Preparation / Synthesis
- Search / Charting
- Pilot screening
- ICT consultation
- Protocol revision / Reporting
- Protocol registration
- Search refinement / Full screening



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Inbox (4,836) - a.neuro.jo@gn

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Neurotechnology in Africa: A Scoping Review



Need training?

Upgrade

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Overview

Review data

Screening

Full text screening

Data extraction

Risk of bias

Blind mode ☒

Screening criteria

🗒️

Stay caught up with review updates

 Edit info

 Invite members

 PRISMA

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Review Info

 Review Title: Neurotechnology in Africa: A Scoping Review

 Review Type: Scoping Review

 Review Domain: Other

 Description: Add Description to your Review!

Data Summary

Imported References

15,498

Add References

Total Duplicates

8,894

Detect Duplicates

Unresolved

8,892

Continue Resolving

☒ Resolved

1

☐ Not Duplicate

☐ Deleted

1

Quick guide

5/5

OSF SCOPING REVIEW REGISTRATION PROTOCOL TEMPLATE

Based on the OSF Generalized Systematic Review Registration Form (van den Akker et al., 2023)

HOW TO USE THIS TEMPLATE Complete all fields before registering on OSF Registries (osf.io/registries). For items not applicable to scoping reviews (e.g., quality appraisal, meta-analysis), write "N/A" and briefly explain why. All fields are mandatory when registering on OSF. PRISMA-ScR checklist items are noted where relevant. For JBI-aligned reviews, upload to OSF first, then submit the OSF link via the JBI Title Registration Form.

SECTION 1: METADATA

[GSRRF-1] Title of Scoping Review

Provide the full title. Clearly indicate this is a scoping review (not a systematic review). PRISMA-ScR: 1.

Neurotechnology in Africa: A Scoping Review

[GSRRF-2] Contributors / Author Team

List all authors/contributors, their institutional affiliations, and contact details for the corresponding author.

Daniella Patron¹, Johanné Marais¹, Mosa Mafalo¹, Sachin Ajodhya¹, Martin Bekker³,
Tasneem Hassem¹, Becky Inkster, Devon Jarvis², Bianca Moffet⁴, Sahba Besharati¹

1. School of Human and Community Development, University of the Witwatersrand

2. School of Computer Science and Applied Mathematics, University of the Witwatersrand

3. School of Electrical and Information Engineering, University of Witwatersrand, Johannesburg

4. School of Public Health

[GSRRF-3] Subjects / Topic Areas

List the key subject areas or disciplines this review covers (e.g., neuroscience, public health, education). These will be used as subject tags on OSF.

Neurotechnology, public policy, governance, ethics, neuroscience, philosophy, neurology, engineering

[GSRRF-4] Tasks and Roles (CRediT)

Describe the expected tasks and roles of each author/contributor, ideally using the Contributor Roles Taxonomy (CRediT). Example roles: Conceptualization, Search Strategy, Screening, Data Extraction, Writing.

“CRedit Roles and Example Research Tasks That Could be Attributed to Them”
https://zenodo.org/records/18421449
Funding acquisition: S. Besharati
Conceptualization: All team members
Search Strategy: J. Marais, D. Patron, M. Mafalo
Project admin and resource management: J. Marais
Data curation: J. Marais, D. Patron, M. Mafalo
Screening: All team members
Data charting: J. Marais, D. Patron, M. Mafalo
Supervision: S. Besharati
Validation: J. Marais, D. Patron, M. Mafalo
Writing (original draft): J. Marais, D. Patron, M. Mafalo
Writing (review and editing): All team members

SECTION 2: REVIEW OVERVIEW & BACKGROUND

[GSRRF-5] Type of Review

Confirm this is a scoping review. State which methodological guidelines or frameworks you are following (e.g., JBI Manual for Scoping Reviews, Arksey & O'Malley framework, PRISMA-ScR). PRISMA-ScR: 1.

This is a Scoping Review, utilizing the PRISMA-ScR framework.

[GSRRF-6] Review Stages

List the planned stages of your review in order. Typical scoping review stages: Preparation → Search → Pilot Screening → Protocol Update (if needed) → Full Screening → Data Charting → Synthesis → Reporting. Include any pilot phases.

Preparation → Search → Pilot Screening → Consulting an Information and Communications Technician → Protocol revision → Registration of protocol → Search → Pilot Screening → Full Screening → Charting → Synthesis of themes → Reporting

[GSRRF-7] Current Review Stage at Registration

Indicate which stage you are at when you freeze this registration. For a prospective pre-registration, this should ideally be 'Preparation' (before searching). Note if this is an update to a previous registration.

Protocol revision

[GSRRF-8] Planned Start Date

Date when the review search is planned to begin, or actual start date if already commenced.

01 April 2026

[GSRRF-9] Planned End Date

Estimated date of completion. Use resources such as PredicTER.org to estimate review duration.

01 October 2027

[GSRRF-10] Background & Rationale

Introduce the topic, explain why this scoping review is needed, summarise relevant existing literature, and describe what this review adds. Note any related existing reviews. PRISMA-ScR: 2 & 3.

Global research in, policy for, and applications of neurotechnology in the age of the fourth industrial revolution are expanding. A handful of international initiatives are collaborating on structuring considerations of neurorights, neural security, and data autonomy of neurotechnology (Bianchi et al., 2018; Greely et al., 2018) into guidelines, with a shared priority of upholding neuroethics integrity in neurotechnological acceleration. Yet, international perspectives on ethical frameworks frequently decenter the input of the Global Majority (Palk, 2022; Robertson et al., 2025). There are hundreds of cultural practices, healthcare systems, and political infrastructures in Africa which can participate meaningfully in building neurotechnology governance frameworks that are socially sensitive and locally implementable. We consider that the success of these frameworks will depend on their relevance for African contexts, in at least three spheres: ethics of neurotechnology, neuroscience of neurotechnology, and governance of neurotechnology. Since there are no unifying articles to date that scope African literature at this intersection, we propose the scoping review of neurotechnology in Africa. We formulate the proposal for the scoping review with the broad questions: what are the foundations of neuroethics in Africa (Matshabane et al., 2024), what are the neuroscientific priorities in Africa (Besharati and Akinyemi, 2023; Donald et al., 2022; Maina et al., 2021) that may look to neurotechnology for solutions now or the near future; and whether policies are currently used (or in development) by African nations to maintain neurotechnological safety.

1. Besharati, S., Akinyemi, R. (2023) Accelerating African neuroscience to provide an equitable framework using perspectives from West and Southern Africa. *Nat Commun* **14**, 8107. <https://doi.org/10.1038/s41467-023-43943-3>
2. Bianchi, D. W., Cooper, J., Gordon, J. A., Heemskerk, J., Hodes, R. J., Koob, G. F., Koroshetz, W. J., Shurtleff, D., Sieving, P. A., Volkow, N. D., Churchill, J. D., & Ramos, K. M. (2018). Neuroethics for the National Institutes of Health BRAIN Initiative. *Journal of Neuroscience*, 38(50), 10583–10585. <https://doi.org/10.1523/jneurosci.2091-18.2018>
3. Donald KA, Maina M, Patel N, Nguemini C, Mohammed W, Abubakar A, Brown M, Stoyanova R, Welchman A, Walker N, Willett A, Kariuki SM, Figaji A, Stein DJ, Ihunwo AO, Daniels W, Newton CR. (2022) What is next in African neuroscience? *Elife*. 22;11:e80488. doi: 10.7554/eLife.80488.
4. Maina, M.B., Ahmad, U., Ibrahim, H.A. *et al.* (2021) Two decades of neuroscience publication trends in Africa. *Nat Commun* **12**, 3429. <https://doi.org/10.1038/s41467-021-23784-8>
5. Matshabane, O. P., Hartford, A., Ewuoso, C., Palk, A. C., Koehly, L. M., Stein, D. J., & Vries, J. de. (2024). Advancing neuroethics in Africa. *South African Journal of Science*, 120(5/6). <https://doi.org/10.17159/sajs.2024/18180>
6. Palk, A. C. (2022). Towards a More Globally Inclusive and Relevant Neuroethics. *Challenges in Medical Ethics: The South African Context*, 157–182. <https://doi.org/10.52779/9781991201959/09>
7. Robertson, L.J., Higgins, N.L., Maier, M.J. *et al.* (2025). Ethical Governance Strategies for the Responsible Innovation of Neurotechnologies: A Scoping Review. *Bioethical Inquiry* 22, 931–948 . <https://doi.org/10.1007/s11673-025-10440-9>

SECTION 3: RESEARCH QUESTIONS & OBJECTIVES

Scoping reviews typically use broad, exploratory research questions rather than the PICOS framework. The PCC framework (Population, Concept, Context) recommended by JBI is commonly used instead.

[GSRRF-11] Primary Research Question(s)

State the specific question(s) this review is designed to answer. These should directly inform your search strategy, inclusion/exclusion criteria, and data charting plan. Consider using the PCC (Population, Concept, Context) framework. PRISMA-ScR: 4.

What is the current evidence base for the past, present, and future of neurotechnology in African populations, research agendas, and policy frameworks?

This informs the choice of our search terms (Section 6.6)

[GSRRF-12] Secondary Research Question(s)

List any additional questions that played a lesser role in shaping the review design. Write N/A if none. PRISMA-ScR: 4.

△ Scoping review note: If this item does not apply, write N/A and briefly explain why.

What applications are researched in the African contexts (e.g., health, industry, science)?

Are there hubs of neurotechnology research publications in Africa?

What themes are present in the collection of publications?
Are ethics, safety and risk discussed in the publications, beyond the ethical clearance for research?

[GSRRF-13] Expectations / Scope Boundaries

Describe any expectations about the literature (e.g., types of sources anticipated, geographic spread, time period coverage). Note any known contextual factors that may influence interpretation. PRISMA-ScR: 3.

We expect that we should find publications not earlier than 1990 which are relevant to the scoping review, as this decade seems to differentiate neurotechnology from broader neuroimaging. Our decision is informed by one particular historical overview (https://neurotech-course.github.io/neurotech_history/) and the milestone papers referenced therein.

From the overview, we include Ogawa et al.'s 1990 paper on fMRI as evolution from MRI (<https://doi.org/10.1073/pnas.87.24.9868>); Chance et al. (1993), Kato et al. (1993), Hoshi et al. (1993), and Villringer et al. (1993) research group papers on fNIRS as evolution from NIRS; FDA approval for deep-brain stimulation to treat Parkinson's in 1997; and brain-computer implants into human brains by the late 1990's

We anticipate that the applications will be focused on health and recovery, work, industry, prosthetics, philosophy, and ethical governance rather than moral and cognitive enhancement. Since we are curious not only about research, but also policy, we will broaden the search to include government white papers, non-governmental organization reports, and industry input documents

SECTION 4: CONCEPTS, POPULATION & CONTEXT (PCC)

Scoping reviews use PCC rather than PICOS. Complete the fields below. If your review does not involve a specific population or intervention, adapt accordingly and note N/A where not applicable.

[GSRRF-14] Main Concept(s) / Outcome(s) of Interest

Describe the core concept(s) this review maps. In PICOS terms, this corresponds to outcomes/main variables. PRISMA-ScR: 10a.

We anticipate that the outcomes of the review will include:

1. Where in Africa various types of neurotechnologies are applied.
2. Which types of neurotechnologies are applied in various countries.
3. What the foundational neuroscientific research focus is, where these technologies are applied (recovery from injury, enhancement, treatment/therapy, curiosity, work, industry and so on).
4. Whether African governance or policy frameworks have been developed for neurotechnology applications.

5. Commentary or opinion pieces which advocate for or against particular best practices in neurotechnology research.

[GSRRF-15] Population / Participants

Describe the population, participants, or phenomena of interest. Include relevant characteristics such as age, condition, setting, or species if applicable. Write N/A if not applicable. PRISMA-ScR: 10b.

△ Scoping review note: If this item does not apply, write N/A and briefly explain why.

Primary research in human participants; reviews on human participant research; opinion pieces or commentary on societal impacts of neurotechnology; policy documents relating to human participants

[GSRRF-16] Context / Setting

Describe any contextual factors or settings relevant to inclusion (e.g., geographic, cultural, clinical, or disciplinary context). Write N/A if not applicable. PRISMA-ScR: 10b.

△ Scoping review note: If this item does not apply, write N/A and briefly explain why.

The flexibility and breadth of our search terms are chosen to scope the full umbrella of what is published. Our inclusion and exclusion criteria will refine the publications to works published in African contexts, across disciplines from neuropsychology to imaging, computational neuroscience or policy - for example.

SECTION 5: TEAM, SOFTWARE & ETHICS

[GSRRF-17] Software & Tools

List all software you will use: reference management (e.g., Zotero, EndNote), screening (e.g., Covidence, Rayyan), data charting, and analysis/synthesis. Include version numbers where known. PRISMA-ScR: 13d.

1. Primary document sharing, protocol drafting, and task delineation; analysis/synthesis of themes and qualitative descriptions: Google Suite
2. Reference management tools: Zotero
3. Storage of resources in a repository: GitHub
4. Screening and charting (often called extraction in systematic reviews)*: Rayyan AI

*note, Rayyan's AI tool will not be used for extracting information from articles unless it is used as supplementary and confirmatory of charting protocols carried out by the research team.

5. Refine search term strategies and sources that guide protocol development: Scopus AI and SciteAI
6. ?Predictive modelling for decision making: Dr Devon Jarvis' tool -

<https://link.springer.com/article/10.1007/s10661-026-15376-0>
<https://github.com/Gavin2417/Machine-Learning-Scoping-Review>

[GSRRF-18] Funding

List all funding sources for everyone involved. If unfunded, state this explicitly. PRISMA-ScR: 25.

This project was supported by the Manufacturing, Engineering and Related Services SETA (Sector Education and training Authorities; merSETA) of South Africa.

[GSRRF-19] Conflicts of Interest

Disclose any potential conflicts of interest. If none, state explicitly. PRISMA-ScR: 26.

None to declare

[GSRRF-20] Overlapping Authorship

Declare whether any member of the review team is an author of studies likely to be included. Describe how this will be managed (e.g., exclusion from screening relevant papers).

In cases where one or many of the review team are authors of a study, a review pair will be allocated to that paper which does not include the author. If the study is included, any extraction will not be completed by the author(s).

SECTION 6: SEARCH STRATEGY

6.1 Sources to be Searched

[GSRRF-21] Information Sources

List all databases and sources to be searched (e.g., MEDLINE, EMBASE, CINAHL, PsycINFO, SCOPUS, Web of Science, grey literature sources, hand searching). PRISMA-ScR: 7.

Web of Science, Google Scholar, Scopus, ProQuest, PsychINFO, EbscoHost, PUBMED, Medline will be searched.
https://www.ieee.org/ will be searched for grey literature sources.
https://proquest.libguides.com/pais - should be covered in ProQuest
https://wils.org/products/political-science-complete/ - should be covered in EBSCO

6.2 Search Strategy

[GSRRF-22] Search Strategy (at least one database)

Provide the full, reproducible search string for at least one key database. Include boolean operators, MeSH/controlled vocabulary terms, and free-text terms. Consider working with a health sciences librarian. PRISMA-ScR: 8.

("governance" OR "management" OR "oversight" OR "regulation*" OR "policy") AND ("risk*" OR "uncertainty" OR "hazard" OR "threat" OR "safety" OR "security" OR "protection" OR "well-being" OR "concern*") AND ("neurotechnolog*" OR "brain technolog*" OR "neural device*" OR "cognitive technolog*" OR "EEG" OR "ERP" OR "Event-related potential*" OR "electroencephalogra*" OR "computeri*ed neuropsychological assessment*" OR "computeri*ed neuropsychological screening*" OR "digital *cognitive assessment" OR "computeri*ed *cognitive assessment*" OR "digital screening tool*" OR "computeri*ed screening tool" OR "fMRI" OR "functional magnetic-resonance imaging" OR "fNIRS" OR "functional near-infrared spectroscopy" OR "eye-tracking" OR "IVR" OR "immersive virtual reality" OR "brain imaging") AND ("application*" OR "monitoring" OR "use" OR "implementation" OR "utili*ation" OR "framework*")

6.3 Search Scope

[GSRRF-23] Date Range

Specify the date range for the search (e.g., 2000 to present). If no date restriction, state this and explain. PRISMA-ScR: 7.

Only studies from 1990-2026 will be included. This is to ensure that more up to date and relevant studies are found. We will still scope the research without date restriction to screen for any foundational studies we may exclude by our 36-year window, and may decide to include seminal policy documents or foundational papers that contextualize neurotechnology in Africa.

[GSRRF-24] Language Restrictions

State whether you will apply language restrictions, and if so, which languages will be included. Justify any restrictions. PRISMA-ScR: 7.

Studies will only be included if in English or if English translations of articles occur. This is to ensure that the researchers are able to understand the article, and no errors in understanding occur, which may impact on analysis of the article and overall application. However, if there are policy documents which are not translated but remain relevant, every effort will be made to catalogue these and name its relevance for neurotechnology in its country of origin.
--

[GSRRF-25] Other Search Limits

Describe any other search limits (e.g., publication type, geographic region, human/animal studies). Justify each restriction. Write N/A if none. PRISMA-ScR: 7.

⚠ Scoping review note: If this item does not apply, write N/A and briefly explain why.

The articles have to be published in online sources described in section 6.1. There are no geographic region publication restrictions - all publications which pass primary searches will be screened for eligibility; the targeted population or policy has to describe an African context. As the funding for the project focuses on neurotechnology and the future of work, studies carried out on animals will be excluded.

SECTION 7: SCREENING & ELIGIBILITY

7.1 Inclusion & Exclusion Criteria

[GSRRF-26] Inclusion Criteria

List all inclusion criteria clearly. Organise by PCC elements (Population, Concept, Context) and study characteristics (e.g., study design, publication type, date). PRISMA-ScR: 6.

Inclusion criteria will be outlined by PCC framework (population, context, concept)

To be included at primary screening, studies must either:

1. mention the use of neurotechnology in an African context, or
2. frame ethical, governance, or safety considerations of neurotechnology in an African context.

Studies will only be included for charting if the research populations are residents of the African continent, or the framework/theory/structure/policy speaks to the African context (All African countries will be accepted).

Neurotechnologies will be defined as any device or mechanism or software used to record, analyse or stimulate electrical activity in the nervous system (Cometa et al., 2022; Müller & Rotter, 2017). Thus, any neurotechnology intervention that meets this definition will be included. Mixed neurotechnology interventions will also be included.

Only studies published between January 1990- January 2026 will be included. This is to ensure that there is a comprehensive understanding of how neurotechnology has been applied in the last 36 years in the African context, including if there have been any updates or innovations. All types of studies will be included regardless of the disorder or condition type, as the goal is to understand the broad use of neurotechnology and its applications in the African context.

Only published studies that are available on online journals will be included. This is to ensure that all included studies can be compared and contrasted digitally. All study designs will be included, and systematic reviews will be filtered for any additional resources which were missed by the database search.

[GSRRF-27] Exclusion Criteria

List all exclusion criteria, with brief rationale for each where helpful. PRISMA-ScR: 6.

Studies will be excluded if they do not utilise participants from African countries or speak to frameworks of safety in African countries, as the goal of this study is to understand the use of neurotechnology/neurotechnologies in Africa. As such, including other populations would not enable us to understand the African context.

Studies will be excluded if they are published before 1990. This is because we are focused on the most relevant and up-to-date use of neurotechnology in this context.

7.2 Screening Process

[GSRRF-28] Screening Levels

Describe the number of screening levels (title/abstract screening, then full-text screening is standard). State whether screening will be conducted independently by two reviewers at each level. PRISMA-ScR: 9.

Databases will be searched by 3 researchers using the search string independently on the mentioned databases. The articles will then be uploaded to Zotero where duplicates will be removed. Then the three researchers will independently download the final articles, screen the title and abstract for relevance. Once these have been screened and consensus has been reached, the researchers will then screen the full texts. Once this is complete, consensus will be reached for the final number of articles. Rayyan AI will be used to streamline process management by importing the articles into the software, where reviewers will still have all executive control and decision-making power

[GSRRF-29] Conflict Resolution

Describe how disagreements between screeners will be resolved (e.g., consensus discussion, third reviewer arbitration). PRISMA-ScR: 9.

Disagreements will be resolved by consensus discussion and third reviewer arbitration.

[GSRRF-30] Pilot Screening Plan

Describe any planned pilot testing of inclusion/exclusion criteria (recommended practice). State the number of records to pilot and how inter-rater reliability will be assessed (e.g., Cohen's kappa). Write N/A if not planned.

⚠ Scoping review note: If this item does not apply, write N/A and briefly explain why.

Pilot testing of search terms and inclusion/exclusion criteria will take place throughout the initial search stages. Preliminary searches will be conducted on a database by changing

search terms/string, date and type of publication by each researcher to ensure that consistent results are found. External assistance from the librarian will be utilised to modify and optimise search string. Inter-rater reliability will be assessed by Cohen's kappa.

SECTION 8: DATA CHARTING

Scoping reviews use 'data charting' rather than 'data extraction'. There is no quality appraisal / risk of bias assessment in standard scoping reviews — write N/A for those fields if using the OSF Generalized Systematic Review template.

[GSRRF-31] Data Charting Form

Describe the data to be charted from each included source. List all variables/fields, including bibliographic information, study characteristics, and content-relevant data. Attach or link your data charting form if available. PRISMA-ScR: 11.

- Data will be synthesised into data tables with the relevant data for the study. Study characteristics will be tabulated according to; author(s), year of publication, country of origin (population), aims/purpose, study context (sample size, methodology/methods) type of Neurotechnology utilised, intervention type/duration, comparator, outcome measures and key findings (content).

[GSRRF-32] Charting Process

Describe who will chart data, whether charting will be done independently by two reviewers, and how disagreements will be resolved. PRISMA-ScR: 11.

The charting process will be conducted independently by pairs of reviewers. Once draft tables have been completed, any disagreements will be discussed as a group and third reviewer may act as mediator.

[GSRRF-33] Pilot Charting Plan

Describe any planned pilot testing of the data charting form. Write N/A if not planned.

⚠ Scoping review note: If this item does not apply, write N/A and briefly explain why.

After information has been agreed upon by both researchers, the researchers will pilot the charting of data. Once complete, the data charting will be compared and refined based on feedback and if any disagreements occur.

SECTION 9: SYNTHESIS & REPORTING

Scoping reviews do not typically conduct meta-analysis or statistical pooling. Write N/A for any synthesis fields that do not apply, with brief explanation.

[GSRRF-34] Synthesis Approach

Describe how charted data will be synthesised and presented (e.g., narrative synthesis, tabular summary, thematic analysis, descriptive mapping). PRISMA-ScR: 13.

Our data charting procedure will be guided by the review's four guiding questions as well as the five anticipated outcomes. It will be synthesised using a mixed-methods approach, using both quantitative descriptive synthesis to map the data and qualitative thematic synthesis. Descriptive and geographic mapping will be utilised in the extraction process to show where in Africa neurotechnology research and application are concentrated, which technologies are applied where, and whether identifiable research hubs exist. Tables, geographic-mapping and publication frequency by country and/or region, the distribution of technology type on the continent, the presence or lack thereof of government frameworks will be explored quantitatively. While qualitative data will include the foundational research focus underlying technologies and the stance of commentary and/or opinion sources through narrative synthesis by theme. Allowing the review to characterise not only in numbers, geographic distribution and the evidence base trajectory, but also the substantive content, application intent as well as the governmental landscape underpinning said evidence.

[GSRRF-35] Quantitative Synthesis

Describe any planned quantitative summaries (e.g., counts, frequencies, descriptive statistics). Write N/A if not applicable.

△ Scoping review note: If this item does not apply, write N/A and briefly explain why.

Quantitative synthesis will include descriptive synthesis of the volume, geographic distribution, and technological type in the evidence base. Simple descriptive analytics will be used to address the review's questions on publication trends, technology types, research hubs, and the governance landscape across the African continent.

- Publication trend over time: annual counts of papers addressing neurotechnology and/or neurorights on the African continent, illustrating growth of the field over time.

- Geographic distribution: country- and region-level frequencies of research and application, presented as a map (e.g., choropleth) to identify potential research hubs and underrepresented areas.
- Cross-tabulation: showing how frequently each technology type is applied along with country/region. Results will be visualised using either a heatmap or stacked bar chart to highlight differences across countries.
- Research focus distribution: frequency of underlying neuroscientific research purpose, presented as a bar chart.
- Governance and policy framework presence: counts of countries with identifiable governance or policy frameworks for neurotechnology, presented as a presence/absence indicator. It can be expanded upon if there is layers to the governance or policy frameworks around neurotechnology.

[GSRRF-36] Qualitative / Thematic Synthesis

If applicable, describe any planned qualitative or thematic synthesis approach. Write N/A if not applicable.

△ Scoping review note: If this item does not apply, write N/A and briefly explain why.

Qualitative synthesis will proceed in two stages. First, thematic analysis will be conducted on the included sources, with the choice between an inductive or deductive coding approach to be finalised once initial familiarisation with the dataset is complete. Coding rigour will be ensured through dual independent coding: a minimum of four coders will code each source, with discrepancies resolved through discussion. Where consensus cannot be reached between the four coders, a fifth coder will be brought in to adjudicate; if disagreement persists, the matter will be escalated to the full review team for a final decision on classification. This staged approach ensures both consistency and a documented audit trail for any contested coding decisions.

Second, coded findings will be synthesised using narrative synthesis, weaving the coded themes and sub-themes into a coherent account of the neurotechnology evidence base within the African context, drawing out how the technology types, research foci, geographic patterns, and governance/advocacy positions identified in the quantitative synthesis interact with and inform one another qualitatively. This narrative will be the primary vehicle for situating quantitative patterns alongside the substantive why and how behind them.

Findings will be presented through three approaches:

- a summary table of themes and quotations,
- narrative synthesis by theme to characterise the African evidence base, and
- evidence gap mapping to highlight underrepresented technologies, regions, or populations.

[GSRRF-37] Reporting Framework

State which reporting guidelines you will follow when writing up results (e.g., PRISMA-ScR). Confirm that a PRISMA-ScR flow diagram will be included. PRISMA-ScR: 17.

The PRISMA-ScR reporting guidelines will be used when writing up results. The PRISMA flow diagram will be included.

SECTION 10: PROTOCOL DEVIATIONS

Complete this section during or after the review to document any deviations from this registered protocol. Transparent reporting of deviations is a key open science practice. If no deviations occurred, state this explicitly in your final report.

[DEV-1] Deviations from Registered Protocol

Describe any deviations from this registered protocol, the reason for each deviation, and the date on which the decision was made. If no deviations, state 'None'.

None currently as of 1 July 2026.

[DEV-2] Updated Registration

If a significant protocol change was made, did you submit an updated registration on OSF? Provide the DOI/link to any updated registration if applicable.

Any updates will be processed within this OSF registration.

SECTION 11: OSF SUBMISSION CHECKLIST

Before submitting to OSF Registries, confirm the following:

- ☐ All mandatory fields completed (or N/A with explanation)
- ☐ Review title clearly identifies this as a scoping review
- ☐ Registration submitted BEFORE database searching begins (for prospective pre-registration)
- ☐ License selected: recommended CC BY 4.0 International
- ☐ OSF project created to house supplementary materials (search strings, charting forms, etc.)
- ☐ If JBI-aligned: OSF link ready to submit via JBI Title Registration Form
- ☐ PRISMA-ScR checklist reviewed and ready to complete post-review

REFERENCES & RESOURCES

Registration form based on:

van den Akker, O.R., Peters, G.J.Y., Bakker, C.J., et al. (2023). Increasing the transparency of systematic reviews: presenting a generalized registration form. *Systematic Reviews*, 12, 170. <https://doi.org/10.1186/s13643-023-02281-7>

Key guidelines for scoping reviews:

JBI Manual for Evidence Synthesis: Scoping Reviews (Section 10): <https://synthesismanual.jbi.global>

PRISMA Extension for Scoping Reviews (PRISMA-ScR): <https://doi.org/10.7326/M18-0850>

OSF Registries: <https://osf.io/registries>

OSF Generalized Systematic Review Registration Form: <https://osf.io/by27q/>

July 2026

Pre-registration of Scoping Review has been published on Open Science Framework (<https://doi.org/10.17605/OSF.IO/RQTNF>)

5 June 2026

("governance" OR "management" OR "oversight" OR "regulation*" OR "policy") AND ("risk*" OR "uncertainty" OR "hazard" OR "threat" OR "safety" OR "security" OR "protection" OR "well-being" OR "concern*") AND ("neurotechnolog*" OR "brain technolog*" OR "neural device*" OR "cognitive technolog*" OR "EEG" OR "ERP" OR "Event-related potential*" OR "electroencephalogra*" OR "computeri*ed neuropsychological assessment*" OR "computeri*ed neuropsychological screening*" OR "digital *cognitive assessment" OR "computeri*ed *cognitive assessment*" OR "digital screening tool*" OR "computeri*ed screening tool" OR "fMRI" OR "functional magnetic-resonance imaging" OR "fNIRS" OR "functional near-infrared spectroscopy" OR "eye-tracking" OR "IVR" OR "immersive virtual reality" OR "brain imaging") AND ("application*" OR "monitoring" OR "use" OR "implementation" OR "utili*ation" OR "framework*")

- ❖ These were chosen as the final search terms to be included in the OSF registration
- ❖ The registration draft was readied and circulated to the team for their input, so that we could register the scoping review on Open Science Framework.

3 June 2026

Meeting with Mosa and Dani

1. ("governance" OR "management" OR "oversight" OR "regulation*") AND ("risk*" OR "uncertainty" OR "hazard" OR "threat" OR "safety" OR "security" OR "protection" OR "well-being") AND ("neurotechnolog*" OR "brain technology" OR "neural devices" OR "cognitive technology" OR "EEG" OR "ERP" OR "Event-related potentials" OR "electroencephalography" OR "neuropsychological assessment" OR "neuropsychological screening" OR "digital cognitive assessment" OR "digital screening tools" OR "fMRI" OR "functional magnetic-resonance imaging" OR "fNIRS" OR "functional near-infrared spectroscopy" OR "eye-tracking" OR "IVR" OR "immersive virtual reality" OR "brain imaging") AND ("application" OR "monitoring" OR "use" OR "implementation" OR "utili*" OR "framework")

pubmed.ncbi.nlm.nih.gov/?term=%28%22governance%22+OR+%22management%22+OR+%22oversight%22+OR+%22regulation%22%29+AND+%22risk%22+OR+%22uncertainty%22+OR+%22hazard%22+OR+%22threat%22+OR+%22safety%22+OR+%22security%22+OR+%22protection%22+OR+%22well-being%22%29+AND+%22neurotechnolog%22+OR+%22brain+technolog%22+OR+%22neural+device%22+OR+%22cognitive+technolog%22+OR+%22EEG%22+OR+%22ERP%22+OR+%22Event-related+potential%22+OR+%22electroencephalogra%22+OR+%22computeri*ed+neuropsychological+assessment%22+OR+%22computeri*ed+neuropsychological+screening%22+OR+%22digital+cognitive+assessment%22+OR+%22computeri*ed+cognitive+assessment%22+OR+%22digital+screening+tool%22+OR+%22computeri*ed+screening+tool%22+OR+%22fMRI%22+OR+%22functional+magnetic-resonance+imaging%22+OR+%22fNIRS%22+OR+%22functional+near-infrared+spectroscopy%22+OR+%22eye-tracking%22+OR+%22IVR%22+OR+%22immersive+virtual+reality%22+OR+%22brain+imaging%22%29+AND+%22application%22+OR+%22monitoring%22+OR+%22use%22+OR+%22implementation%22+OR+%22utili%22+OR+%22framework%22%29

NIH National Library of Medicine
National Center for Biotechnology Information

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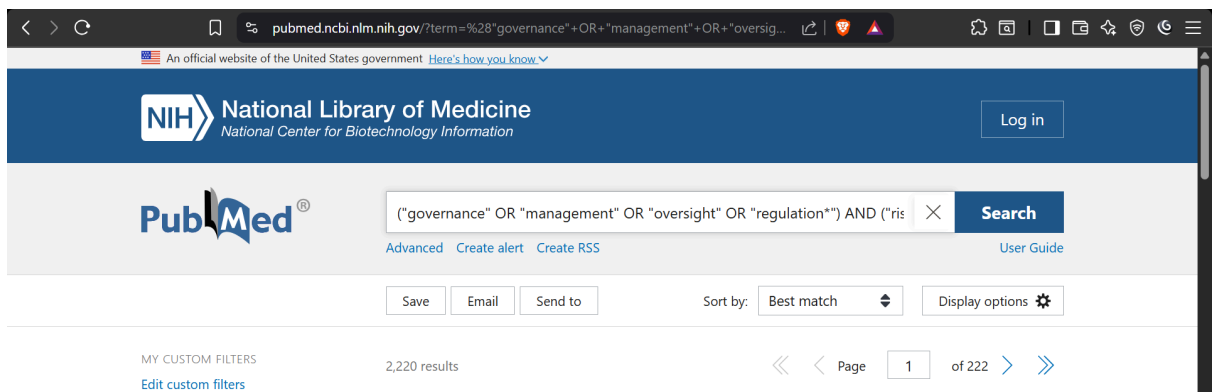
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Advanced Create alert Create RSS User Guide

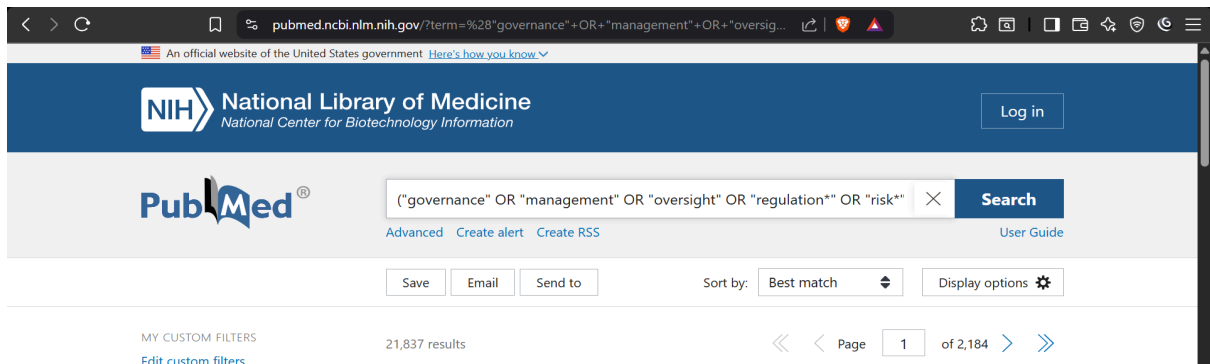
Save Email Send to Sort by: Best match Display options

MY CUSTOM FILTERS Edit custom filters 2,259 results Page 1 of 226

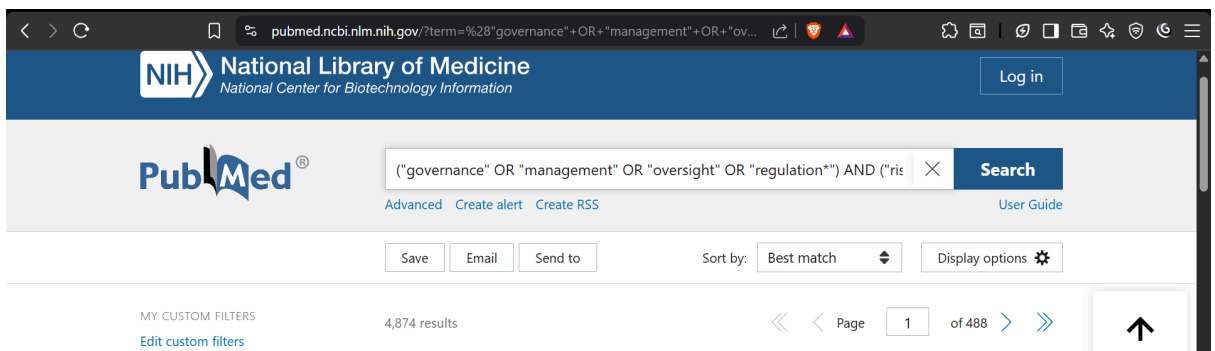
2. ("governance" OR "management" OR "oversight" OR "regulation*") AND ("risk*" OR "uncertainty" OR "hazard" OR "threat" OR "safety" OR "security" OR "protection" OR "concerns") AND ("neurotechnolog*" OR "brain technology" OR "neural devices" OR "cognitive technology" OR "EEG" OR "ERP" OR "Event-related potentials" OR "electroencephalography" OR "neuropsychological assessment" OR "neuropsychological screening" OR "digital cognitive assessment" OR "digital screening tools" OR "fMRI" OR "functional magnetic-resonance imaging" OR "fNIRS" OR "functional near-infrared spectroscopy" OR "eye-tracking" OR "IVR" OR "immersive virtual reality" OR "brain imaging") AND ("application" OR "monitoring" OR "use" OR "implementation" OR "utili*" OR "framework")



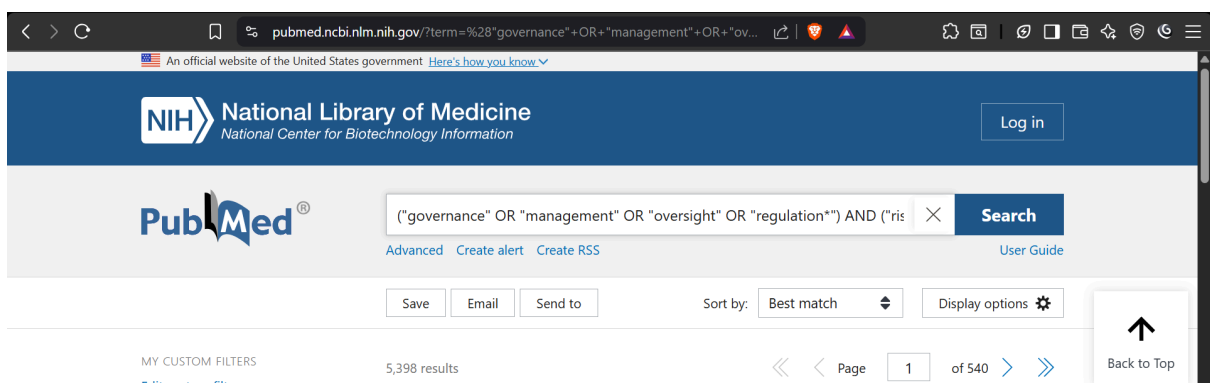
3. ("governance" OR "management" OR "oversight" OR "regulation*" OR "risk*" OR "uncertainty" OR "hazard" OR "threat" OR "safety" OR "security" OR "protection" OR "concerns") AND ("neurotechnolog*" OR "brain technology" OR "neural devices" OR "cognitive technology" OR "EEG" OR "ERP" OR "Event-related potentials" OR "electroencephalography" OR "neuropsychological assessment" OR "neuropsychological screening" OR "digital cognitive assessment" OR "digital screening tools" OR "fMRI" OR "functional magnetic-resonance imaging" OR "fNIRS" OR "functional near-infrared spectroscopy" OR "eye-tracking" OR "IVR" OR "immersive virtual reality" OR "brain imaging") AND ("application" OR "monitoring" OR "use" OR "implementation" OR "utili*" OR "framework")



4. ("governance" OR "management" OR "oversight" OR "regulation*") AND ("risk*" OR "uncertainty" OR "hazard" OR "threat" OR "safety" OR "security" OR "protection" OR "well-being") AND ("neurotechnolog*" OR "brain technology" OR "neural devices" OR "cognitive technology" OR "EEG" OR "ERP" OR "Event-related potentials" OR "electroencephalography" OR "neuropsychological assessment" OR "neuropsychological screening" OR "digital cognitive assessment" OR "digital screening tools" OR "fMRI" OR "functional magnetic-resonance imaging" OR "fNIRS" OR "functional near-infrared spectroscopy" OR "eye-tracking" OR "IVR" OR "immersive virtual reality" OR "brain imaging" OR "computeri*") AND ("application" OR "monitoring" OR "use" OR "implementation" OR "utili*" OR "framework")



5. ("governance" OR "management" OR "oversight" OR "regulation*") AND ("risk*" OR "uncertainty" OR "hazard" OR "threat" OR "safety" OR "security" OR "protection" OR "well-being" OR "concern*") AND ("neurotechnolog*" OR "brain technology" OR "neural devices" OR "cognitive technology" OR "EEG" OR "ERP" OR "Event-related potentials" OR "electroencephalography" OR "neuropsychological assessment" OR "neuropsychological screening" OR "digital cognitive assessment" OR "digital screening tools" OR "fMRI" OR "functional magnetic-resonance imaging" OR "fNIRS" OR "functional near-infrared spectroscopy" OR "eye-tracking" OR "IVR" OR "immersive virtual reality" OR "brain imaging" OR "computeri*") AND ("application" OR



"monitoring" OR "use" OR "implementation" OR "utili*" OR "framework")

19 May 2026

I began doing some research in the previous week about where we can begin participating in global ethics conversations to familiarize ourselves with the existing networks and identify potential collaborative partnerships. Two stand-out options are:

1. Global Forum for Bioethics Research: This is an opportunity to attend either as participant or presenter and applicants from LMICs can be offered grants to support their attendance. This is an application-based forum rather than an open-to-the-publics forum. <https://www.gfbr.global/forum-meetings/>
2. The Ethics Research Lab has a Wellcome Research Development Programme in which they run webinars, catalyst grants, or shorter-term residencies for researchers. <https://health.uct.ac.za/ethics-lab/projects/wellcome-research-development-programme#>

I also did some work on the search terms, and propose these as the terms to be shared with the group during the meeting on 20 May 2026:

- ("Africa*" OR ... list by name all African countries?) AND ("governance" OR "management" OR "oversight" OR "regulation*") AND ("risk" OR "uncertainty" OR "hazard" OR "threat") AND ("risk*" OR "safety" OR "security" OR "protection" OR "well-being") AND ("neurotechnology" OR "brain technology" OR "neural devices" OR "cognitive technology" OR "EEG" OR "ERP" OR "Event-related potentials" OR "electroencephalography" OR "digital cognitive assessment" OR "digital screening tools" OR "fMRI" OR "functional magnetic-resonance imaging" OR "fNIRS" OR "functional near-infrared spectroscopy" OR "eye-tracking" OR "IVR" OR "immersive virtual reality") AND ("application" OR "use" OR "implementation" OR "utilization")

6 and 7 May 2026

During these two days, I began the mapping process for important information to include in the scoping review. This started with the process of translating the recommendations documents into an outline for what the team can collaborate on ahead of our registration, listed here [Neurotechnology at work: an African Scoping Review](#)

Additionally, I developed a template using Claude AI to summarize from Open Science Framework's requirements for registering a scoping review here

[OSF_Scoping_Review_Registration_Template.docx](#)

29 April 2026

On 28 April, I finalized the briefing document for the Wits Faculty of Health Sciences information technician at the library to assist us with search term development. This is linked here: [W systematic search protocol and pilot search results - search form_ .docx](#)

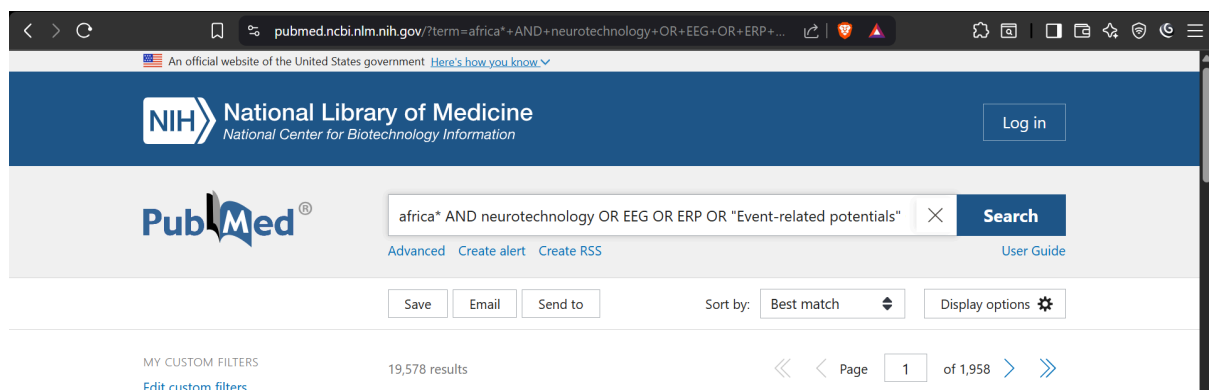
On 29 April, we met with Mr Devind Peter and he offered us insights into several Wits-offered platforms that support scoping reviews and tips for strengthening search terms. These outcomes from the meeting are detailed here:

[W Jo Marais - 29 April 2026 session on search strategies .docx](#) . Unfortunately, many of these services are only available with an institutional access email and we are awaiting these to be distributed to the team.

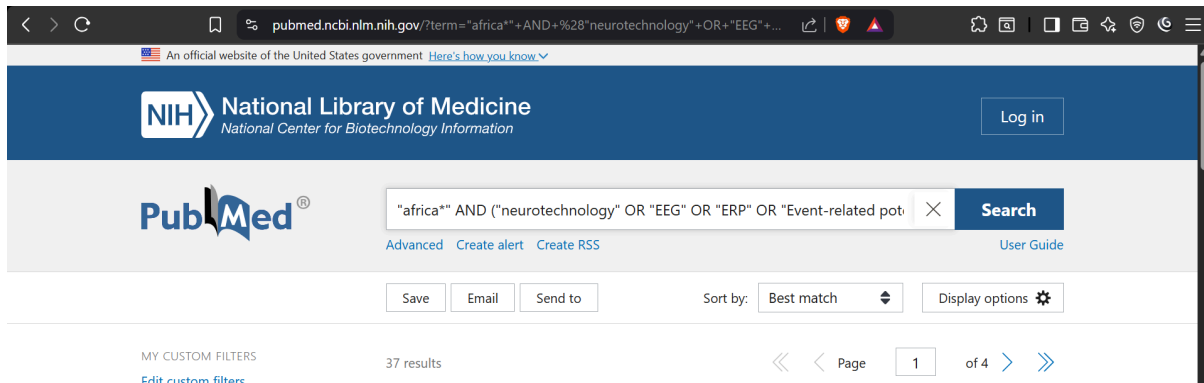
1. (africa*) AND (neurotechnology)) OR (EEG)) OR (ERP)) OR ("Event-related potentials")) OR (electroencephalography)) OR ("digital cognitive assessment")) OR ("digital screening tools")) OR (fMRI)) OR ("functional magnetic-resonance imaging")) OR (fNIRS)) OR ("functional near-infrared spectroscopy")) OR (eye-tracking)) OR (IVR)) OR ("virtual reality")) AND (ethics)

... 18 000 results

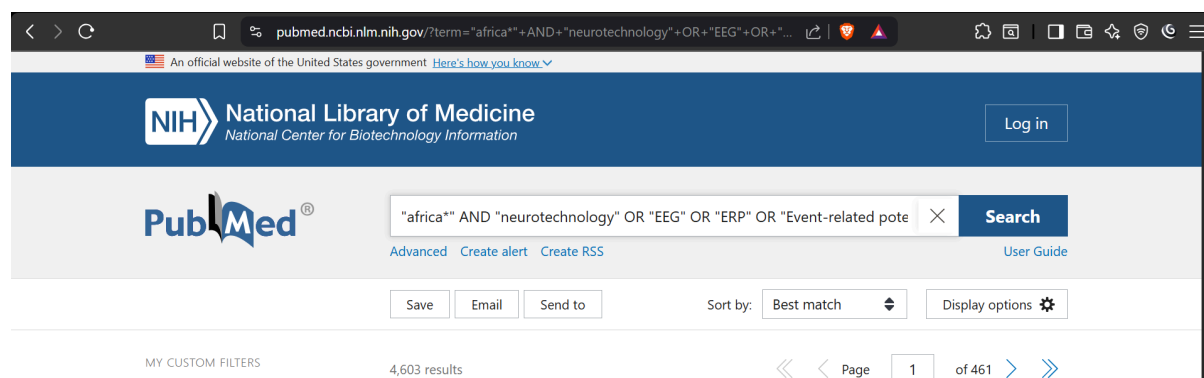
2. africa* AND neurotechnology OR EEG OR ERP OR "Event-related potentials" OR electroencephalography OR "digital cognitive assessment" OR "digital screening tools" OR fMRI OR "functional magnetic-resonance imaging" OR fNIRS OR "functional near-infrared spectroscopy" OR eye-tracking OR IVR OR "virtual reality" AND ethics



3. "africa*" AND ("neurotechnology" OR "EEG" OR "ERP" OR "Event-related potentials" OR "electroencephalography" OR "digital cognitive assessment" OR "digital screening tools" OR "fMRI" OR "functional magnetic-resonance imaging" OR "fNIRS" OR "functional near-infrared spectroscopy" OR "eye-tracking" OR "IVR" OR "virtual reality" AND "ethics")



4. "africa*" AND "neurotechnology" OR "EEG" OR "ERP" OR "Event-related potentials" OR "electroencephalography" OR "digital cognitive assessment" OR "digital screening tools" OR "fMRI" OR "functional magnetic-resonance imaging" OR "fNIRS" OR "functional near-infrared spectroscopy" OR "eye-tracking" OR "IVR" OR "virtual reality" AND "ethics"



Scopus Ai – search string from meeting with Mr Peter

- ("governance" OR "management" OR "oversight" OR "regulation") AND ("risk" OR "uncertainty" OR "hazard" OR "threat") AND ("safety" OR "security" OR "protection" OR "well-being") AND ("neurotechnology" OR "brain technology" OR "neural devices" OR "cognitive technology") AND ("application" OR "use" OR "implementation" OR "utilization")

14 April 2026: 07h28 to 08h49

PubMed:

When the search terms are individually typed into the search phrasing with ADD, AND, OR rather than added with our own grammar into the query box, the grammar changes to say this...

((((((((((((((africa*) AND (neurotechnology)) OR (EEG)) OR (ERP)) OR ("Event-related potentials")) OR (electroencephalography)) OR ("digital cognitive assessment")) OR ("digital screening tools")) OR (fMRI)) OR ("functional magnetic-resonance imaging")) OR (fNIRS)) OR ("functional near-infrared spectroscopy")) OR (eye-tracking)) OR (IVR)) OR ("virtual reality")) AND (ethics)

18 343 results

All historical publications

Google Scholar:

When I type the following directly into the search bar rather than using the advanced search...

(Africa*) AND (neurotechnolog*) OR (ERP) OR (event-related potentials) OR (EEG) OR (electroencephalography) OR (digital cognitive assessments) OR (digital screening tools) OR (IVR) or (virtual reality) OR (eye-tracking) OR (fMRI) OR (fNIRS) OR (functional magnetic resonance imaging) OR (functional near-infrared spectroscopy) AND (ethics)

... I yield 89 results.

Closing up for this process today:

- ❖ As these are too much or too few, I returned to the group to refine terms again.
- ❖ It was suggested we reach out to Mr Devind Peter, an information technologist, at Wits Health Science Library, to assist with search terms.

10 April 2026: 13h51 to 14h38

PubMed

All Fields:

(Africa*) AND ((neurotechnolog*) OR (ERP) OR (event-related potentials) OR (EEG) OR (electroencephalography) OR (digital cognitive assessments) OR (digital screening tools) OR (IVR) or (virtual reality) OR (eye-tracking) OR (fMRI) OR (fNIRS) OR (functional magnetic resonance imaging) OR (functional near-infrared spectroscopy)) AND (ethics)

193 results

All historical publications

Google Scholar:

3 results

Closing up for this process today:

❖ Shared with the wider group for input

2 April 2026 12h20 to 12h27 and 3 April 13h41 to 13h57

I am repeating searches across databases according to the workflow from 31 March 2026, using the following search terms:

(Africa* AND neurotech*) AND (Brain-computer interfac* OR (BCI*) OR (neuro-prosthe*) OR (EEG) OR (Electroencephalogra*) OR (fMRI) OR (Functional magnetic resonance imaging) OR (electrocorticogra*) OR (ECoG) OR (VR digital testing) OR (VR) OR (Virtual reality) OR (Wearable headset*) OR (Neurofeedback) OR (ERP) OR (patch-clamp*) OR (DBS) OR (Deep brain stimulation) OR (Transcranial magnetic stimulation) OR (TMS) OR (tDCS) OR (Focused ultrasound) OR (FUS) OR (Brain implants) OR (neural implants) OR (Transcranial direct current stimulation) OR (Cochlear-implan*) OR (Vagus nerve stimulation) OR (Brain-machine systems) OR/AND Ethics OR risk OR safety OR applications

PubMed

The search yielded over 6 million results!

If I repeat the search filtering through MeSH terms, I draw just under 2 million outputs. There may be value in applying each type of neurotechnology filter on the database search.

Closing up for this process today:

❖ Shared with the wider group for input

1 April 2026 15h13 to 16h19

After reviewing Sahba's email in response to the searches of 31 March 2026, I have drafted this list of search terms

Africa* AND
neurotech* AND
Brain-computer interfac* OR
BCI* OR
neuro-prosthe* OR
EEG OR
Electroencephalogra* OR
fMRI OR
Functional magnetic resonance imaging OR
electrocorticogra* OR
ECoG OR

VR digital testing OR
Virtual reality OR
Wearable headset* OR
Neurofeedback OR
ERP OR
patch-clamp* OR
DBS OR
Deep brain stimulation OR
Transcranial magnetic stimulation OR
TMS OR
tDCS OR
Focused ultrasound OR
FUS OR
Brain implants OR
Neural implants OR
Transcranial direct current stimulation OR
Cochlear-implan* OR
Vagus nerve stimulation OR
Brain-machine systems OR
AND
Ethics OR risk OR safety OR applications

These search terms were informed by the following resources:

<https://post.parliament.uk/neurotechnology-what-is-it-and-how-is-it-used/>

<https://www.technologynetworks.com/neuroscience/articles/neurotechnology-358488>

<https://doi.org/10.7551/mitpress/15173.001.0001> - *Neuroethics: The Implications of Mapping and Changing the Brain*, Walter Glannon

Closing up for this process today:

- ❖ Shared with the wider group for input

31 March 2026: 08h30 - 10h00

1. I set up a Covidence-registration for the scoping review
2. I began the search across repositories/databases starting with PubMed using the steps below:

PubMed 1:

- Click on Advanced Search under the search bar
- Change terms in query box to be "Title/abstract" and type agreed search terms with logic, then click "Add" so that the query box is Africa[Title/Abstract] OR African[Title/Abstract] AND neurotechnology[Title/Abstract] AND safety[Title/Abstract] OR ethics[Title/Abstract] OR risk[Title/Abstract] OR applications[Title/Abstract]
- Then click search. This yielded 4,369,653 results

PubMed search 2:

- Edit the query box to read ((Africa OR African) AND (Neurotechnology)) AND (ethics OR safety OR risk OR applications)
- 32 results

PubMed search 3:

- Edit the query box to read ((Africa OR African) AND (Neuro technology)) AND (ethics OR safety OR risk OR applications) as suggested by PubMed
- 729 results

I decided to stick with neurotechnology, rather than “neuro technology” since this is not commonly the way neurotechnology is referred to. On the PubMed search 2 criteria above:

I selected Save; Selection: All results; Format: PubMed; Create file

3. I moved on to EBSCOhost, though it seems I need an institutional account to sign in.
4. I moved on to Google Scholar:

Google Scholar 1:

Click on left hand panel to open Advanced Search, then Find articles with all the words: Neurotechnology AND Africa*, and at least one of the words: ethics OR Risk OR safety OR application, by Title search

- 2 results which are the same publication
5. Next was Web of Science
 - I ran into some trouble with finding the Advanced Search bar within my WoS profile. I wonder if this is because I do not use an institutional email address anymore?

Closing up for this process today:

- ❖ I uploaded the references I found into the Covidence setup.
- ❖ Updated Dani and Sahba
- ❖ Will search some databases which do not require institutional email addresses